Question	Answer	Mark	Mark scheme	Additional guidance
22	21	P1	for a relevant probability, eg P(green) = $\frac{x}{2x+3}$ or P(blue) = $\frac{x+3}{2x+3}$	the number of green and blue pens could be $x - 3$ and x or equivalent probabilities must be in an algebraic form in a single variable
		P1	for a relevant product, eg. $"\frac{x}{2x+3}" \times "\frac{x-1}{2x+2}"$ or $"\frac{x+3}{2x+3}" \times "\frac{x+2}{2x+2}"$ **OR** $\left("\frac{x}{x+3}"\right)^2 + \left("\frac{x+3}{2x+3}"\right)^2 = \frac{27}{75}$	This is an exception using replacements. No further credit is available
		P1	forms an appropriate equation, eg. $"\frac{x}{2x+3} \times \frac{x-1}{2x+2}" + "\frac{x+3}{2x+3} \times \frac{x+2}{2x+2}" = \frac{27}{55}$	
		P1	(dep P3) process to reduce equation to $ax^2 + bx + c = 0$ eg. $x^2 - 25x + 84 = 0$	
		P1	process to solve quadratic equation eg. $(x - 21)(x - 4) = 0$	
		A1	cao	